Microcontroller Skill Share Activity Sheet

This is an assisted exploration of micro-controllers using an “RP2040 Zero” microcontroller.

The plan for today is to see how easy it is to set up a new microcontroller and make the on-board LED light up with different colors. If all goes well, you’ll get to do a little bit of programming and see some of the changes that you can make. Don’t worry! We’re here to help.

Quick Start Instructions

This is the abbreviated version of the instructions for getting set up. If you need more details, see the following pages or ask for help. You’ll be using an RP2040 Zero (Raspberry Pi microcontroller) for this exploration.

1.) Plug the device into your computer with a USB C cable. Press the RESET and BOOT buttons together then release the buttons one at a time starting with the RESET button and followed by the BOOT button.
2.) Copy the uf2 file (adafruit-circuitpython-waveshare_rp2040_zero-en_US-10.1.4.uf2) from the skill-share-system folder to the RPI-RP2 drive.
3.) Copy the contents of the skill-share-examples folder (Just the contents, not the folder itself) to the CIRCUITPY drive.
4.) Open https://code.circuitpython.org/ in Chrome or Explorer.
    a.) Select the USB interface that the device is plugged into
    b.) Select CIRCUITPY as the root directory.
    c.) Make sure both the Editor and Serial windows are open.
5.) Load the “code.py” file into the editor.
6.) Proceed to the section labeled “Writing and Running Programs” in this document.

Getting Set Up - In detail

Step 1: Get a microcontroller and take it out of its package. Notice that there is a usb connector and two white buttons on the microcontroller. One button is labeled “boot” the other is labeled “reset”.

Even though the controller is just about the size of a postage stamp, it’s actually a pretty powerful little computer.

Step 2: Open a file explorer window. Find the USB C cable connected to your computer and while holding the “boot” button down plug the cable into the microcontroller, then release the button.   After a few seconds, you should see a new drive called “RPI-RP2” show up on your system. This is the micro-controller…it looks like a USB stick or “Thumb Drive”. Copying files onto this drive is how you will get your programs onto the micro-controller.

Step 3: There’s a folder on your desktop called “Skill Share-CIRCUITPY”. Copy the file named “adafruit-circuitpython-waveshare_rp2040_zero-en_US-10.1.4.uf2” to the USB drive. This is the “circuit-python” software. In a few moments, circuit python will be installed. The “RPI-RP2” drive will go away and be replaced by one called “CIRCUITPY”.

Step 4: Find the folder named “Skill Share-Examples”. Open it and copy all the files you find there to the CIRCUITPY drive on. Now you’re ready to do some programming.

Opening the Development Environment - In detail

With circuit-python installed, the microcontroller is programmed with a computer language called “python”. Python is a popular language because you can do some very sophisticated things with it, but it is also easy to get started. The exercises we suggest today are going to be on the easy side.

Circuit python has a very simple model for running programs. If you put a file called ‘code.py’ onto the CIRCUITPY drive the controller will run it as soon as you create it, or if you make changes and save them to the file.

However, there’s an even nicer way to interact with the system.

Step 0: Use Chrome or Explorer to open this link to https://code.circuitpython.org/ and click on the “USB” button.

Step 1: click on “Connect to Device”, click on the USB port with a name like “Circuit Python CDC control (COM13) - Paired” then click the “CONNECT” button. (The number may be different.)

Step 2: click on the “Select New Folder” button and use the file browser to navigate to the CIRCUITPY drive, hit “select” and then click “Use CIRCUITPY”

Step 3: On the bottom left on the browser window there are two buttons one says “Editor” the other says “Serial”. If either button is grey, click on it so that it turns purple.

Step 4: Finally, click on the “Open” button near the top of the window and open the file “code.py” from the CIRCUITPY drive.

Writing and Running Programs.

With circuit-python installed, the microcontroller is programmed with a computer language called "Python". Python is a popular language because you can do some very sophisticated things with it, but it is also easy to get started. The exercises we suggest today are going to be on the easy side.

If you've followed all of the steps above, there should be a red light on the controller that is blinking on and off. The light should be changing once every ½ second.

The program that is running is displayed on the left half of the browser. This is the editor window. The editor shows the lines of the program with numbers to the left. These numbers aren't part of the program, they're just there to help us identify what line we're talking about. The interesting part of the program is the last four or five lines, the ones number 22-25. For now you can ignore all the stuff above that. Just think of lines 1-18 as setup that you won't need to change for now.

The last four lines are set up to run in order and then repeat in order forever. (We call that a loop.) Each line is responsible for doing one thing. Lines 22 and 24 both use the "led.fill" function to set the color of the LED on the board. The led.fill function takes a parameter which is the color of the light. [Setting the color to BLACK is the same thing as turning the LED off.]

The function "sleep" tells the micro-controller to pause for some number of seconds. In this case the pause is for ½ (.5) seconds.

To make changes to the program, just click on the "save" button in the editor and the updated program will start running in a second or two.

Here are some things to try:

1. Try changing program so that the blinking goes faster? [Hint: you'll have to make the same change in two places]
2. Try changing the program so that the light blinks alternating RED and GREEN instead of RED and BLACK.
3. Can you add a couple of lines so that the lights change in a 3 color pattern instead of two colors? [hint: you can find a list of pre-defined colors in lines 14-18. You can also try defining your own colors if you want something more.]

If it doesn't work after you make a change:
On the right side of the development window (next to the editor) is a "serial console". There will be messages in that window that can help you figure out what the problem is. Our apologies, but figuring out the specific problems is more than we can describe in this document.

Next Steps

Useful Links:
Circuit Python: https://circuitpython.org/board/waveshare_rp2040_zero/